

Data Mining :-

- It is nothing but a process of extracting or finding information from huge vol. of data.
- primary goal is to extract knowledge & insights that can aid in decision-making.

Types :-

1) Classification :-

involves categorizing data into predefined labels based on its characteristics.

for eg. classifying emails as a spam or non-spam based on content.

2) Clustering :-

Groups similar data points without predefined categories.

for eg. you have a dataset that contains info about your purchase from the shop.

Through clustering, algorithm can group same purchase behaviours among you & other customers, thus help businesses to get target customers.

3) Regression :-

predicts numerical value based on historical data.

for eg. predicting sales figures based on various influencing factors.

4) Association:-

identifies relationships & association betⁿ variables

for eg.

Discovers that customers who buy product A are likely to buy product B as well

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Applications:- C-FIR

- 1) Customer segmentation:- eg. in clustering ✓
- 2) Fraud Detection - unusual patterns, anomalies.
- 3) Image classificatⁿ:-
- 4) Recommendation Systems

9

Challenges SPCR

1) Data Quality:

accuracy & correctness of data impact mining results.

Inaccurate, incomplete data → lead to misleading patterns

2) Computational complexity:

Analyzing large datasets requires substantial computational power & time.

3) Privacy concerns:

extract^g knowledge from data may involve sensitive info. ensuring privacy with regulations is crucial to maintain Trust.

4) Selection of algorithm

choosing right alg → challenging
diff alg + suitable for diff types of data.

Q Basic build^g blocks or components to construct Data Mining process.

1) Data cleaning - address^g errors, miss^g value.

2) Data preprocessing - raw → suitable format
involves normalizatⁿ, discretizatⁿ to standardize data.

3) Pattern discovery:- apply dm algorithms.
involves; cluster^g, classificatⁿ, regression

4) Evaluation - assess^g model's performance
measures such as accuracy, precision, recall, f1 score.

5) Interpretation → understand^g mined patterns
- allow stakeholders to make informed decision.

Q Advantages - DM task primitive

- Scalability
- Flexibility
- Interpretability
- Versatility

Data	Information
- Raw facts, symbols or observat ⁿ	- processed, organized & meaningful data
- unprocessed & unstructured	- processed & presented in structured format
- serves as foundation of inf.	- provides insights, knowledge & decision-making
eg. Temp. reading (eg) 25°C	- Weather report indicates, "25°C & sunny"

Data Warehouse:-

✓ It is a technology that collects that the data from various sources within organization to provide meaningful business insights. The huge amt of data comes from multiple sources / places such as Marketing or finance.

Data Warehouse is used for analysis of data rather than transaction processing.

✓ Therefore, Databases store current, & detailed data and focuses on transactional processing. But D. Warehouse store historical & summarized data.

✓ Data W. uses denormalized star or snowflake schema for query performance

While database uses, normalized schemas

Key-features of Data W.

T I N S.

1) Time-Variant :-

Stores historical data, allowing for time-based analysis & trend identification

2) Integratⁿ

Combines data from various sources into a unified, consistent format.

3) Non-Volatile :-

Once data is loaded, it does not change, ensuring consistency for analysis.

4) Subject-Oriented :-

Organized around specific subjects or themes to support decision making

9 / Need :- Conqueran

- Efficient Reporting

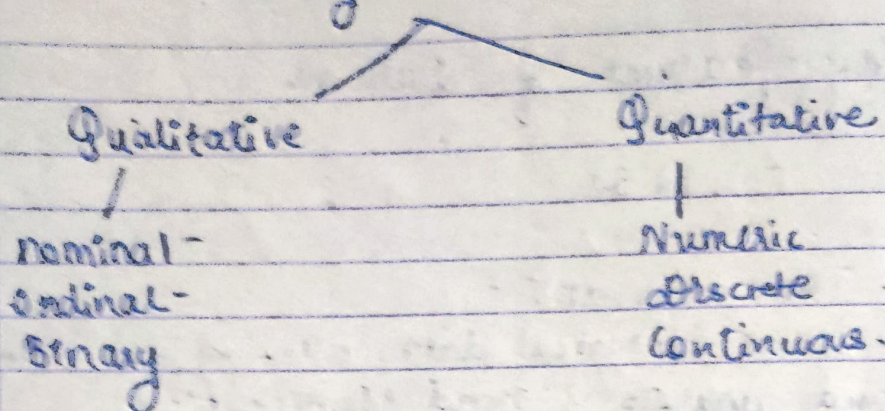
Conqueran

- Complex Queries

- Historical Analysis

- Data Consolidation

Data Mining attributes:-



Nominal → cannot compare.

+ represent category.

+ no order.

Eg. colors → black, red, ...

Ordinal → have meaningful sequence.

Attribute	values
Grade	A, B, C,
Percentage	101, 801, 901.

Binary:

two values.

Eg. yes-no, false-true

Symmetric

Gender - female

Male

Asymmetric

Cancer detector - Two Yes/No

Numeric - measurable.

Eg. Temp.

Discrete:

- has finite values
- can be numerical / categorical

eg Attribute	Values
Pincode	412207, 411030
Profession	Teacher, chef, student.

Continuous

- has infinite no. of values
- can be many values betⁿ 2 numbers.

Attribute	Values
Height	5.1, 5.2, 5.6, 6.1
Weight	50.26, 25.10, 50.50, 50.21

OLTP

- Online transaction processing
- main obj to perform processing
- uses insert, update & delete operations
- Responds faster
- Tables r normalized
- uses original data
- characterized by large no. of short online transaction

OLAP

- Online analytical processing
- main obj to perform analysis.
- uses SELECT operation
- Responds Slower
- Tables r not normalized
- OLTP is source of data.
- characterized by large volume of data.

Fact Table

- purpose - to store quantitative data
- content - numeric, measurable, data such as sales, quantity
- typically larger in size due to detailed data
- Connected to multiple dimension tables -
- changes frequently as new facts r added

Dimensional Table

- store descriptive attributes
- descriptive data like product, time, location
- smaller in size as it contains reference data -
- connected to specific fact through key.
- changes less freq. as it holds reference data -

Types of Data Mining:-

① Relational Database

The relational database is a collection of multiple datasets formally organized by tables, records, columns from which data can be accessed without having to recognize database table.

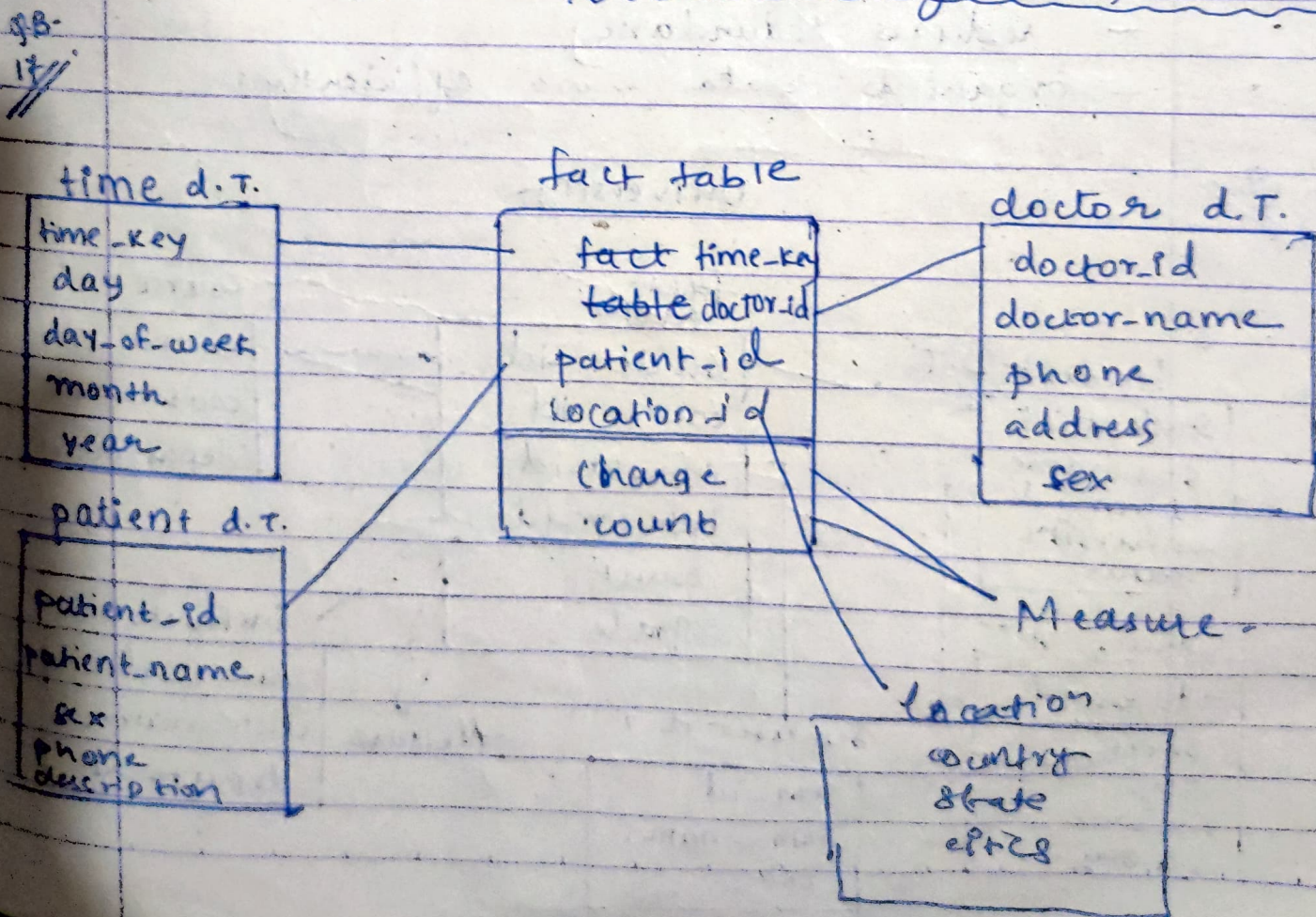
- used for searchability & reporting

② Data Repositories:-

- refers to destination for data storage.
- a grp of db., where org. has kept various kind of inf.

Star Schema:-

- A star schema is a database schema in Data warehousing where there is a central fact table connected to multiple dimensional tables through foreign key, forming $\star$ like structure.
- A fact table consists of numerical measures like facts, such as sales, or revenue.
- while dimension tables has descriptive attributes like time, locatⁿ, products,
- Foreign key relationships connect primary key of F.T To the correspond^g Primary key of D.T.
- It supports complex analytical processes.



Snowflake Schema :-

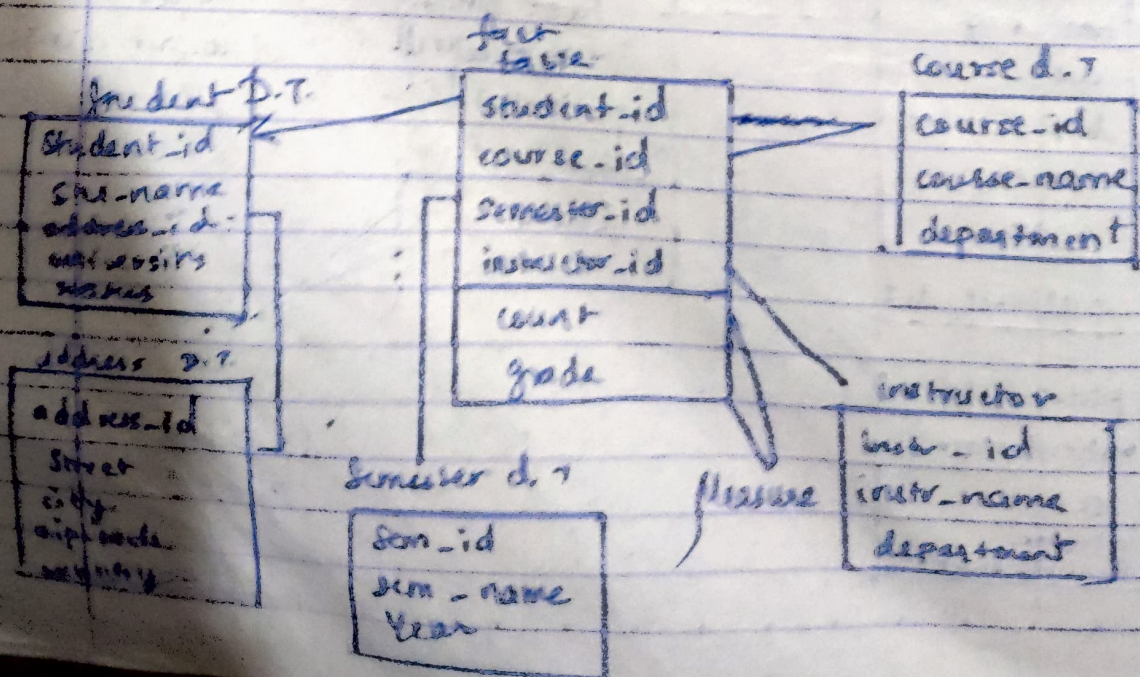
- It is like star schema, is a database in data warehousing for organizing & managing large datasets.
- In contrast to a schema, snowflake extends normalization process by further decomposing dimension table into sub-dimension.
- This result in a shape resembling a snowflake rather than a star.

Structure :- Central fact table surrounded by d.T, similar to star schema.

- But then undergo normalization, where d.T. are decomposed into sub-dimensions.
- reduces redundancy
 - organizes data more efficiently

Q18

University



① LAP operatⁿ:-

① Roll-Up

⇒ Roll up → involves aggregating data from a detailed level to a higher, summarized level.

- Its helpful for obtaining a more generalized overview of dataset, enabling higher level analysis & decision making

Eg:- Aggregating daily sales data into monthly or yearly totals.

② Slice & Dice:-

- Slicing is like focusing on a specific part of your data,
- Like a selecting a particular dimension or attribute & analyze that data.

Eg:-

If u have sales data for all products, slicing would be looking at sales of a particular product category.

Dice:- Similar to slicing but involves looking at specific intersections of data.

Eg:-

If u have sales data for products & region. Dicing would be looking at sales of specific product in particular region

ROLAP

MOLAP

HOLAP

→ relational db

→ Multidimensional

→ Hybrid.

→

→ more storage space required

→ efficient storage using cubes

→ balanced storage.

→ slower for complex queries.

→ faster for analytical queries.

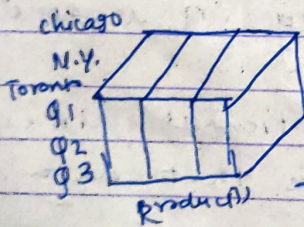
→ combines benefit of both

→ More flexible

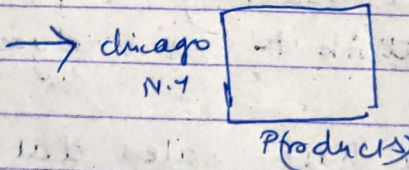
→ requires fixed structure with cubes

→ offers balance of flexibility & structure.

Slice



slice on time dimension.



UNIT - 3.

1) Proximity Measures in Data Mining:

- - also known as similarity or dissimilarity measures, are used to quantify degree of similarity or dissimilarity between objects in dataset.
- These measures are crucial in various data mining tasks as clustering, classification, and anomaly detection.
- Here are some p.m.:-

1) Euclidean Distance:-

- widely used
 - calc. straight-line distance between two points
- $$d(p, q) = \sqrt{\sum_{i=1}^n (p_i - q_i)^2}$$
- two points.
between n attributes.

2) Manhattan Distance:-

- calc dist between two points by summing absolute diff. between their ~~dist~~ co-ordinates along each dimension.

$$d(p, q) = \sum_{i=1}^n |p_i - q_i|$$

3) Cosine Similarity:-

- calc. cosines of angle between two vectors
- used in text mining & recommendation systems.
- ranges from -1 to 1.

$$\text{Cosine Similarity}(p, q) = \frac{p \cdot q}{|p| \cdot |q|}$$

Pearson Correlation Coefficient :-

- measures quantifies the linear correlation betⁿ two variables
- used for feature-selection
- ranges from -1 to 1.
 - 1 $\rightarrow$ perfect -ve linear relationship.
 - +1 $\rightarrow$ perfect +ve "
 - 0 $\rightarrow$ no linear relationship.

$$r_{p,q} = \frac{\sum_{i=1}^n (p_i - \bar{p})(q_i - \bar{q})}{\sqrt{\sum_{i=1}^n (p_i - \bar{p})^2 \sum_{i=1}^n (q_i - \bar{q})^2}}$$

$\bar{p}, \bar{q} \rightarrow$ means of p & q .

Q2. Dissimilarity of Nominal attributes & Ordinal Attributes.

$\Rightarrow$ Nominal Attributes :-

- $\rightarrow$ represent categories or labels without any order or sequence.
- $\rightarrow$ Eg. colors, genders, types of cars.
- $\rightarrow$ Dissimilarity betⁿ N.A. means measuring how diff. or similar two categories are without considering any order.

Let's take one example :-

Suppose, we have a dataset of fruits with a "Color" attribute, and we're comparing two fruits, one is "Apple" with color "Red" and another is "Banana" with color "Yellow".

The dissimilarity betⁿ these two fruits based on color is simply 1 if they have different colors & 0 if they have same color.

Ordinal Attributes:-

- also represent categories, but they have a clear order or ranking associated with them
for eg.: education levels like "High School", "Bachelor's Degree", & "Master's Degree" form an ordinal attribute. because there's a clear order from lower to higher education

If we compare two individuals, one with "Bachelor's Degree" & other with "High School" education, dissimilarity betⁿ them is higher compared to two individuals with "B.D." & "M.S."

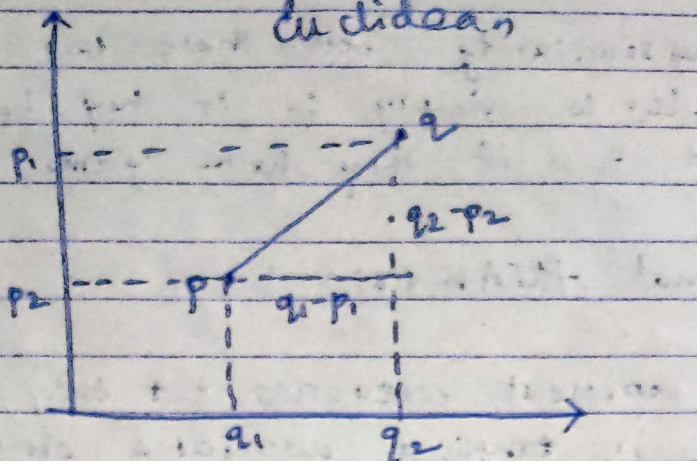
This is bcz the order matters - there's a large gap in educatⁿ level betⁿ "H.S." & "B.D." compared to "B.D." & "M.D."

Eclidean $\rightarrow$...
formula^s Representatⁿ.

For two points $P(x_1, y_1)$ and $Q(x_2, y_2)$ in a two-dimensional space, E.D is,

$$d(P, Q) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

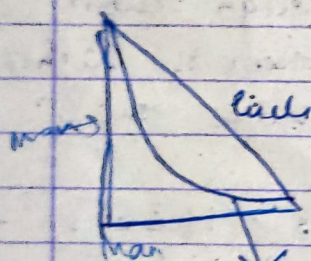
Euclidean



$$d(p, q) = \sqrt{(q_1 - p_1)^2 + (q_2 - p_2)^2}$$

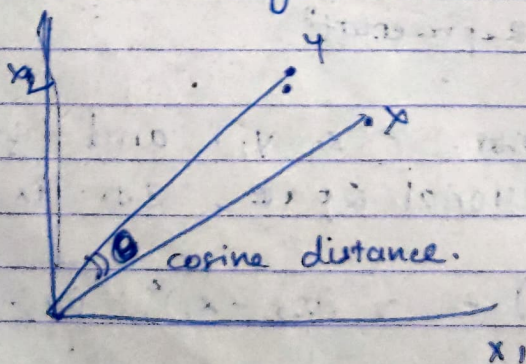
Manhattan Dist

$$d(p, q) = |x_2 - x_1| + |y_2 - y_1|$$



Minkowski

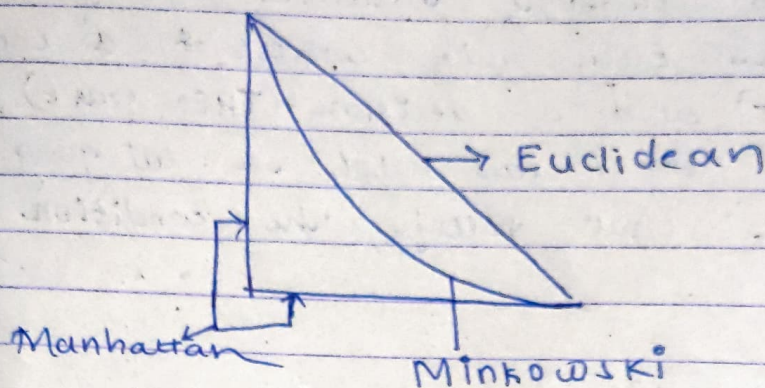
Cosine Similarity



$$d(x, y) = \frac{x \cdot y}{|x| |y|}$$

Minkowski dist-

$$\left[\sum_{i=1}^n |x_i - y_i|^p \right]^{1/p}$$



Different types of similarity techniques
Same as first quest answers.

add 1 more measure there:- & here too.

Minkowski Distance :-

Distance or dissimilarity between two numeric attributes is measured by this in N-dimensional space.

$$d(p, q) = \sqrt[p]{(x_2 - x_1)^p + \dots}$$

$$\left[\sum_{i=1}^n |x_i - y_i|^p \right]^{1/p}$$

It is basically generalization of Euclidean & Manhattan distance.

Minkowski when $p=1 \rightarrow$ Manhattan
when $p=2 \rightarrow$ Euclidean

* Rule-based Classification using IF-THEN Rules for classification.

⇒ type of classificatⁿ method that uses IF-THEN rules to classify instances into different categories. Each rule consist of a condition (IF part) and a decision (THEN part), which specifies the class label or category for instances that satisfy the condition.

Workⁿ

1) Rule Generatⁿ:-

process begins with generating set of rules from training data. Each rule typically corresponds to a specific pattern or combination of feature values that are indicative of a particular class.

2) Rule evaluation:-

The generated rules are evaluated based on their predictive ~~and~~ accuracy or other performance metrics using a validation dataset. Rules may be pruned or refined to improve performance.

3) Rule Applicatⁿ:-

Once the rule set is finalized, it can be applied to classify new instances. Each instances is evaluated against conditions specified in rules, and correspond^g decision or class label is assigned based on rule that is satisfied.

4) Conflict Resolution:-

In some cases, multiple rules may apply to the same instances, leading to conflicts. Various conflict resolution strategies can be used to handle such situations, such as selecting rule with highest confidence or rule with most specific conditions.

Classification

- predicts class or category of an instance
- target var is categorical (discrete)
- evalⁿ metrics are accuracy, precision, recall, F1-score, ROC-AUC
- outputs discrete labels
- eg. spam detection, disease diagnosis, sentiment analysis

Regression

- predicts a continuous numerical value
- target var is continuous (numerical)
- evalⁿ metrics MSE, R-Squared, Mean absolute error
- output continuous values.
- eg. house price prediction, sales forecasting, stock price prediction.

Q1/ Classification & Regression in Predictive Analysis

- class & Reg r two crucial techniques in predictive analysis used to make predictions on input data. They belong to supervised learning, where algorithm learns from labeled data to make predictions on unseen data.

Classification:-

- The goal is to categorize data into predefined classes or categories. It's used when output variable is categorical. Here's how it works:-

1/ Training Phase:-

The alg is provided with labeled dataset where each data instance is associated with a class label.

2/ Model Building:- (M.B)

During M.B, alg learns patterns & relationships betⁿ input features & target classes. Common classificatⁿ alg. include decision trees, random forest, support vector machine (SVMs), Neural networks

3/ Prediction Phase:-

The trained model is used to make predictⁿ on new data instance by classifying them into one of predefined classes.

example:-

consider a spam email classification task. The input features could be various char. of an email (e.g. sender, subject, body content) & the target var is binary (spam or not). The classification alg. learns from labeled eg. of spam & non-spam emails to build a model that can predict whether new, unseen emails are spam or not.

Regression:-

Regression is another predictive modeling technique used when the output variable is continuous or numerical.

Here's how it works:

1. Training phase:-

Similar to classification, in training phase, the algorithm is provided with a labeled dataset where each data instance has a numerical target value.

The alg learns from data to build a model that can predict target value for new, unseen instances.

2. Model Building:-

During model build, alg learns relationship betⁿ input features & continuous target var.

3. Predictⁿ Phase:-

The trained model is used to make predictⁿ on new data instance by estimatⁿ target value based on input features.

Eg:-

Consider a real estate price prediction task. The input features could be various char of a house (eg. size, no of bedrooms, locatn), & target var is price of the house.

The reg alg. learns from historical data on house prices to build a model that can predict price of new, unseen house based on their features

* Concept of Entropy & Informatⁿ Gain.

⇒ The concept of entropy & information gain are fundamental in decision tree algorithm, particularly in the context of feature selection & splitting criteria.

Entropy:-

entropy is the measure of randomness or uncertainty in a dataset. In the context of decision trees, entropy is used to quantify the impurity of a dataset before & after splitting it, based on a particular feature.

Mathematically, Binary classification problem (two classes, such as 'yes' and 'no') the entropy of a dataset is calc as,

$$\text{Entropy}(S) = -P_{\text{pos}} \log_2(P_{\text{pos}}) - P_{\text{neg}} \log_2(P_{\text{neg}})$$

Where:-

- P_{pos} → proportion of positive (or 'yes') instances
- P_{neg} → proportion of negative (or 'no') instances

→ The entropy ranges from 0 to 1.

A low entropy value indicates dataset is pure. while high entropy value indicates high level of uncertainty or impurity.

Information Gain:-

It is measure used to decide which feature to split on at each node of a decision tree.

It quantifies the effectiveness of a feature in reducing uncertainty (or entropy) about the target variable after split.

The I.G. of a feature A w.r.t a dataset S is calc. as diff betⁿ entropy of dataset before & after splitting on feature A:

$$IG(S, A) = Entropy(S) - \sum_{v \in \text{values}(A)} \frac{|S_v|}{|S|} \times Entropy(S_v)$$

S → Entropy before split

$\text{value}(A)$ → possible values of A.

S_v → subset of instances of S

$|S|$ & $|S_v|$ → no. of instances in S & S_v .